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A modelling approach for kinematic equivalent mechanism and rotational transmission error of RV reducer

Yuhu Yang*, Guocheng Zhou, Le Chang, Gang Chen

Key Laboratory of Mechanism Theory and Equipment Design, Ministry of Education, Tianjin University, Tianjin 300350, China

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ABSTRACT

This paper presents a method for the analytical error modelling of RV reducer concerning its over-constrained structure, which explicitly establishes the relations between the original errors and the transmission precision. According to the structural characteristic, i.e., the rotation and revolution of the cycloidal gear share three crankshafts, an equivalent multi-loop mechanism of the reducer is achieved through replacing higher pairs by lower pairs, which decouples the over-constrained relations among the multi-crank actuation. And the error model of the reducer is thus developed by the loop incremental method, in which the manufacturing and assembly errors are represented by the linkage length errors in the equivalent mechanism. Finally, the proposed model is validated by the test results of prototypes, and the influence of original errors is respectively investigated. The outcomes provide a new idea for error modelling and components accuracy design of RV reducer and other precision gear systems.

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1. Introduction

RV reducer, characterised by advantages of high precision, rigidity, and efficiency, has been widely applied in the fields of industrial robot joints, aerospace, machine tools, etc. [1], and tremendous efforts have been devoted towards the investigation of cycloid drives to meet requirements in high-precision transmission fields, including cycloidal tooth profile design [2–4], modification schemes [5–7], stress calculation [8–10], efficiency prediction [11–13], dynamic behaviours [14–16], and the transmission error analysis [17–37].

Throughout the references, studies of RV reducer have always been focused on the error modelling and precision analysis since the rotational transmission error (RTE) is the main performance index that directly reflects the positioning accuracy of robots. And they can be classified into two parallel tracks, i.e., single-stage cycloid drive and entire machine of RV reducer.

Along the first track, in the early stage, Blanche and yang [17, 18] established an analytical backlash model of single-stage cycloid drive with machining tolerances, and concluded that both backlash and torque ripple change periodically with the crank angle, which provided an effective means for accurately calculating transmission error. Ivanović et al. [19] developed a comprehensive mathematical model of trochoidal gearing with clearances, which can be applied to the design of gerotor pumps and cycloid reducers. The above-two researchers regarded errors as the inner equidistance of theoretical tooth profile in their error modelling researches. Zhang et al. [20] established a model to predict the multi-tooth contact conditions and to determine the optimal tooth profile with the consideration of semi-analytical load distribution. Sun et al. [21, 22] pro-

* Corresponding author.
E-mail address: yangyuhu@tju.edu.cn (Y. Yang).

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Nomenclature

O_2O_3,$O_2'O_3'$,$O_2''O_3''$	the eccentricity of the crankshafts
O_1	the rotation centre of the sun gear
O_2,O_4	the geometric centres of the planetary gear and cycloidal gear
N_1,N_2	the curvature centres of involute tooth profiles
k_c	the contact point of the k-th pin and cycloidal gear
C_k	the curvature centre of cycloidal tooth profile
O_k	the centre of the k-th pin
θ_1,θ_2,θ_3,θ_4,θ_5,θ_H	the rotation angles of the sun gear, planetary gear, crankshaft, cycloidal gear, pin gear, and carrier
z_1,z_2,z_4,z_5	the tooth numbers of the sun gear, planetary gear, cycloidal gear, and pin gear
θ_{ai}, θ_{Hi}	the rotation angles of each side of the parallelograms
i	the sequence of the parallelogram
θ_{b1}, θ_{b2}, θ_n, θ_ρ, θ_k	the rotation angles of the virtual linkages O_1N_1,N_1N_2,O_2N_2,C_kO_k, and O_4C_k
α'	the working pressure angle of the involute gear
(x_c, y_c), (x_k, y_k), (x_{O4}, y_{O4})	the coordinates of C_k,O_k, and O_4
θ_p	the angle between the x-axis and O_1O_k
$l_x\boldsymbol{e}^{j\theta x}$	the position vector of the linkage l_x
l_{Hi}, l_{ai}, l_R	the length of the carrier, eccentricity of the crankshafts, and pitch circle radius of the pin gear
l_{b1}, l_n, l_{b2}, l_ρ, l_k	the lengths of the virtual linkages O_1N_1,N_1N_2,O_2N_2,C_kO_k, and O_4C_k
θ_v	the rotation angle of the virtual crankshaft
g_1	the speed ratio of the planetary gear to carrier
g_3	the speed ratio of the cycloidal gear to crankshafts
g_2, g_4	the speed ratio of opposite sides of the parallelograms
Δl_x, $\Delta\theta_x$	the length error and rotation angle error of the corresponding linkage l_x
$\Delta\theta_H$	the rotational transmission error of entire machine
f_1, f_3	the RTE caused by original errors of the involute and cycloid-pin gear transmission subsystems
f_{2i}, f_{4i}	the RTE resulted by original errors of the crankshafts when they work as the input and output subsystems
$\Delta\theta_{b1}$	the input angle error
(E_H, β_H)	the position deviation of holes on output disc
(E_c, β_c)	the position deviation of holes on cycloidal gear
(E_a, β_a)	the eccentricity error of crankshafts
(E_v, β_v)	the centre error of cycloidal gear
Δl_R	the pitch circle radius error of pin gear
δ	the error of cycloidal tooth profile
Δr	the error of pin radius
AP	the accumulative pitch error of pins
(E_{b1}, β_{b1})	the geometrical eccentric error of pitch circle of sun gear
(E_{b2}, β_{b2})	the geometrical eccentric error of pitch circle of planetary gear
ρ_c	the curvature radius of the contact point
(A_o, γ_o)	the assembly error of output disc
k_1	the short width coefficient
m	the module of the involute gear
r	the pin radius
I	the speed ratio of entire machine
f	the frequency of the output error
f_0	the rotation frequency of the output shaft

posed a refined tooth contact analysis model incorporating different modification schemes based on discrete tooth profiles, and indicated by experiment results that the negative isometric together with negative offset was the best combination to reduce transmission error and contact force. Also, Wang et al. [23] proposed a multi-tooth contact model and a transmission error model by dividing the contact area into several differential elements, and concluded that the error mean and its amplitude are respectively determined by modification scheme and meshing period. To obtain more findings, scholars have also deepened relevant studies on different influential factors, such as pin-hole clearances and eccentricity errors [24], tooth profile and pitch errors [25], as well as turning-arm bearing clearances [26]. In addition, Lin et al. [27] introduced a

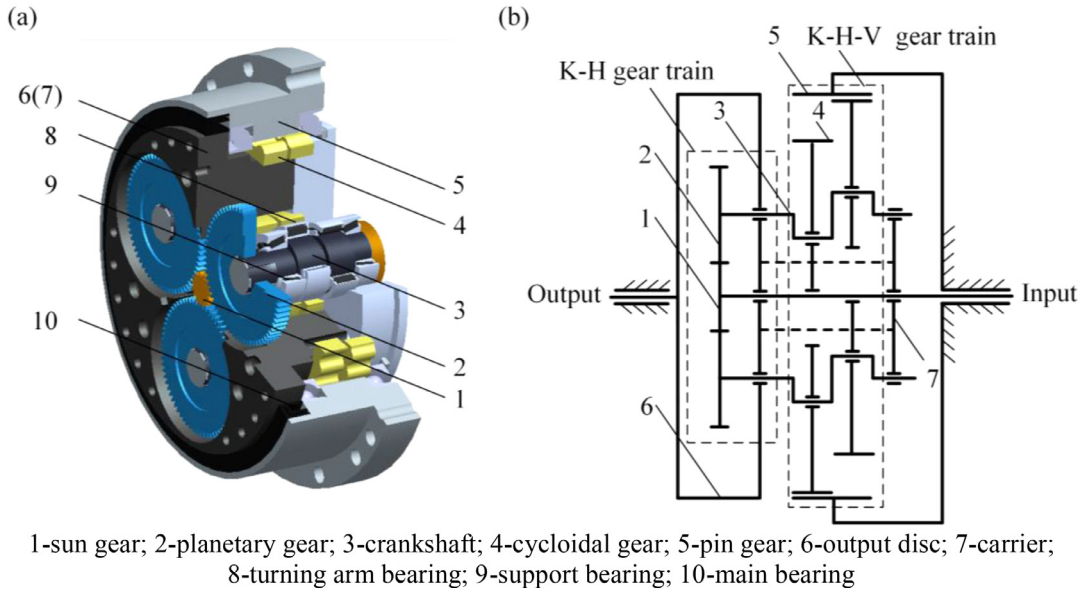


Fig. 1. Structure of RV reducer, (a) CAD model and (b) schematic diagram of mechanism.

method for kinematic error analysis and tolerance design, and optimized the tolerances of parameters with the objective of minimizing manufacturing costs, which provided some meaningful references for parameter design and tolerance allocation.

As the parallel session, the researchers along the second track mainly concentrate on the modelling of RV reducer with consideration of clearances, manufacturing and assembly errors. The early representative research was reported by T. Hidaka et al. [28-30], and they established an accurate and systematic model by employing equivalent spring method to qualitatively analyse the transmission precision of the whole reducer through the numerical solutions. He and Shan [31] established a transmission error model of RV reducer by dynamic sub-structure method and implemented relevant test validation. Meng et al. [32] respectively presented transmission error models of the positive and negative drive based on the static analysis of actual meshing process between gear pairs with frictions considered. Chu et al. [33] produced a selective assembly method to obtain the high-precision reducer from the low accuracy components through analysing the influence of assembly process and manufacturing errors on the backlash. Han and Guo [34] analysed the global error sensitivity to nonlinear and uncertainty factors by Sobol' method, and found out the main factors affecting the transmission accuracy. Furthermore, studies with respect to dynamic accuracy test have also been carried out, such as the tooth profile manufacturing error measurement [35], the error tracing via the test data filtering [36], and vibration suppression by designing feedback system [37]. The aforementioned literatures provided valuable insights for evaluating and improving the accuracy of the transmission system. among the current researches of the single-stage cycloid drive and RV reducer, the former may not be very beneficial for the error modelling of RV reducer due to neglecting the shared multi-crank coupling and their feedback errors; The latter, such as the representative [28-30], established the RTE model by introducing the compatibility equations, considering the over-constrained three crankshafts actuation. Nonetheless, numerical results may have little contribution to indicating the influence of original errors on the RTE due to the lack of explicit expressions. Obviously, explicit expressions are valuable for guiding the tolerance design of the components to obtain the high-precision reducer. And the above exhausted literature review shows that little work was reported on this aspect.

To address the above-mentioned aspect, this paper aims to derive a complete and analytical RTE model of RV reducer. Different from existing studies, the method starts with the investigation of the over-constrained topological structure. Then, it utilizes an equivalent mechanism to decouple the over-constrained relations among the multi-crank actuation rather than deriving the compatibility equations. The significance of this paper is to provide a new idea for establishing the explicit error model of RV reducer by converting its error modelling into the analysis of the equivalent mechanism.

The remainder of this article is organized as follows, after this introduction, the structure and transmission principle of the reducer are introduced in Section 2. Next, the instantaneous kinematic equivalent model is proposed in Section 3. Then, an analytical RTE model is developed in Section 4. Subsequently, the effectiveness of the model is verified by the test results of prototypes in Section 5, before conclusions are drawn in Section 6.

2. Structure and transmission principle

RV reducer is a two-stage closed compound gear train, as the CAD model and schematic diagram of mechanism shown in Fig. 1. The high-speed stage is a K-H type differential gear train, consisting of sun gear 1, three planetary gears 2, and

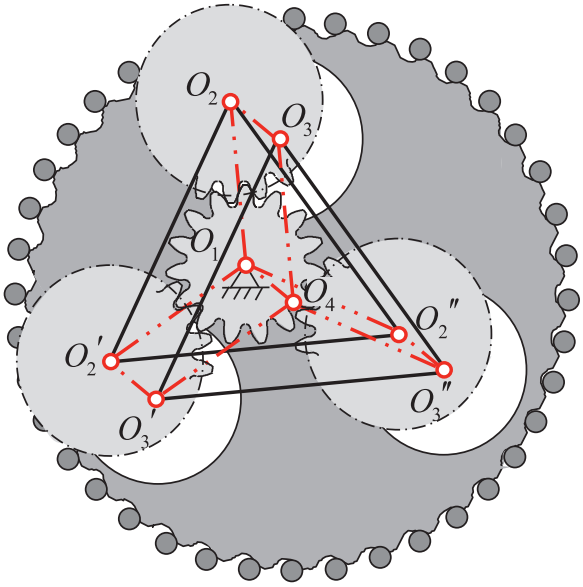


Fig. 2. Equivalent parallelograms.

carrier 7. The low-speed stage is a K-H-V type planetary gear train, including three identical eccentric placed crankshafts 3, two symmetrically arranged cycloidal gears 4, pin gear 5, and output disc 6. The turning arms are connected to the corresponding holes of the cycloidal gears by revolute pairs, and the turning arm bearings 8 are installed in the joints. The output disc and carrier are fixed as one component, and they are connected to the both ends of each crankshaft by revolute pairs with support bearings 9, respectively. A main bearing 10 is installed between the output disc and the shell of the reducer.

When the sun gear rotates, the three planetary gears and crankshafts are driven to rotate around their own axis, and the revolution of the cycloidal gears is actuated via the rotation of the crankshafts. As for the cycloidal gears, through meshing with the pins, they rotate in the opposite direction from their own revolution. Meanwhile, their rotation, as the input, is just fed back to the K-H differential gear train through the carrier. That means the crankshafts are common components of the input and output of the cycloidal gears, thereby accomplishing a closed planetary transmission.

3. Kinematic equivalent mechanism

In order to establish the analytical RTE model with explicit equations for RV reducer, this section intends to first study its instantaneous kinematic equivalent mechanism according to the structural characteristics, and follows the kinematic analysis of the equivalent mechanism to support the error modelling of the reducer in the next section.

3.1. Crankshafts

Three crankshafts with the same eccentric direction (O_2O_3,$O_2'O_3'$, and $O_2''O_3''$) are shown in Fig. 2. O_1 is the rotation centre of the sun gear. O_2 and O_4 are the geometric centres of the planetary gear and cycloidal gear, respectively. O_2O_3 is the eccentricity of the crankshafts.

The crankshafts drive the cycloidal gear to revolve around O_1 with the radius O_2O_3, so the distance from O_4 to O_1 is constantly equal to O_2O_3. Assume that a virtual crankshaft is connected between O_1 and O_4, the motion of the crankshafts can be equivalent to three parallelograms with a shared adjacent side O_1O_4, as depicted with the red double dot dash line in Fig. 2.

3.2. Gear pairs

Based on the method of replacing higher pairs by lower pairs, the equivalent virtual linkages of the tooth profiles are described in Fig. 3.

K-H gear train: N_1N_2 is the internal common tangent of base circles of the two gears, and N_1 and N_2 are the curvature centres of involute tooth profiles, respectively. By the replacement of involute tooth profiles with lower pairs, O_1N_1, N_1N_2, and N_2O_2 are thus regarded as the virtual linkages, and N_1 and N_2 are the virtual hinge centres.

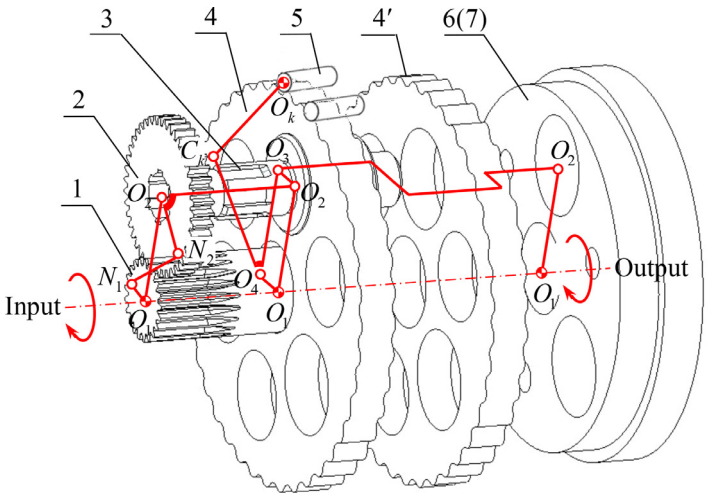


Fig. 3. Lower pairs replacing higher pairs of tooth profiles.

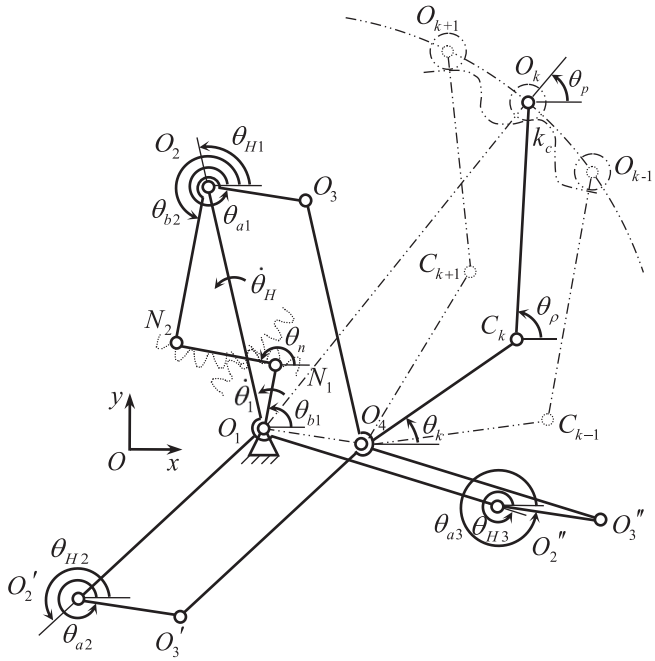


Fig. 4. Instantaneous kinematic equivalent mechanism.

K-H-V gear train: Suppose that the cycloidal gear contacts with the k-th pin at the point of k_c, and the cycloidal tooth profile curvature and pin centres are C_k and O_k, respectively. The cycloidal tooth profile is therefore replaced by the virtual linkages O_4C_k and C_kO_k only with low pairs, and their lengths change with the contact point.

Followed by the above analysis, the instantaneous kinematic equivalent mechanism of the gear train is developed, as illustrated in Fig. 4, and it is analysed to be a multi-loop mechanism.

3.3. Kinematic equations

To develop the explicit RTE expressions of RV reducer, it is necessary to establish the kinematic equation of each virtual link in the instantaneous kinematic equivalent mechanism, as shown in Fig. 4.

(1) Rotation angle of the crankshaft

Referring to Fig. 4, assume that rotation angles of the sun gear, planetary gear, crankshaft, cycloidal gear, pin gear, and carrier are θ_1,θ_2,θ_3,θ_4,θ_5, and θ_H, respectively, and then the speed ratio of the gear train is formulated as

$$i_{12}^{H} = \frac{\theta_1 - \theta_H}{\theta_2 - \theta_H} = -\frac{z_2}{z_1},\ i_{45}^{3} = \frac{\theta_4 - \theta_3}{\theta_5 - \theta_3} = \frac{z_5}{z_4} \tag{1}$$

Given the structure of the gear train, $\theta_3 = \theta_2$, $\theta_4 = \theta_H$, and $\theta_5 = 0$. Substituting them into Eq. (1) yields

$$\theta_H = \frac{z_1(z_5 - z_4)}{z_2 z_5 + z_1(z_5 - z_4)}\theta_1,\ \theta_3 = -\frac{z_4}{z_5 - z_4}\theta_H \tag{2}$$

where z_1,z_2,z_4, and z_5 are tooth numbers of the sun gear, planetary gear, cycloidal gear, and pin gear, respectively.

(1) Rotation angles of the virtual linkages

Parallelograms: The initial position is defined as the virtual crankshaft O_1O_4 coinciding with x-axis in the positive direction, and then rotation angles of each side are expressed as

$$\theta_{ai} = 2\pi - \theta_3,\ \theta_{Hi} = \theta_H + \frac{\pi}{2} + \frac{2\pi}{3}(i-1) \tag{3}$$

where i denotes the sequence of the parallelogram, and $i = 1$–3.

Involute gears: Note that the equivalent mechanism $O_1N_1N_2O_2$ rotates synchronously with the carrier, thus rotation angles of the virtual linkages O_1N_1, N_1N_2, and O_2N_2 are respectively represented as

$$\theta_{b1} = \theta_{H1} - \alpha',\ \theta_n = \theta_{H1} - \alpha' + \frac{\pi}{2},\ \theta_{b2} = \theta_{H1} - \alpha' + \pi \tag{4}$$

where α' is the working pressure angle of the involute gears.

Cycloid-pin gear: When the cycloidal tooth profile contacts with the k-th pin, rotation angles of the virtual linkages C_kO_k and O_4C_k are separately given by

$$\theta_\rho = \arctan\frac{y_k - y_c}{x_k - x_c},\ \theta_k = \arctan\frac{y_c - y_{O4}}{x_c - x_{O4}} \tag{5}$$

where (x_c, y_c), (x_k, y_k), and (x_{O4}, y_{O4}) are the coordinates of C_k,O_k, and O_4, respectively, and they can be calculated by

$$\begin{cases} x_k = l_R\cos\theta_p \\ y_k = l_R\sin\theta_p \end{cases},\ \begin{cases} x_{O4} = l_a\cos\theta_a \\ y_{O4} = l_a\sin\theta_a \end{cases} \tag{6}$$

where θ_p is the angle between x-axis and O_1O_k, and it is expressed as

$$\theta_p = (k-1)\frac{2\pi}{z_5}(k = 1, 2, 3, \cdots, z_5) \tag{7}$$

(1) Kinematic equations of the mechanism

Fig. 5 demonstrates the closed-loop vector model of the mechanism. Assume that the position vector of the linkage l_x is represented as $l_x\boldsymbol{e}^{j\theta x}$, and then kinematic equation of each loop is obtained as

$$l_{b1}\mathbf{e}^{j\theta_{b1}} + l_n\mathbf{e}^{j\theta_n} = l_{H1}\mathbf{e}^{j\theta_{H1}} + l_{b2}\mathbf{e}^{j\theta_{b2}} \tag{8}$$

$$l_v\mathbf{e}^{j\theta_v} + l_k\mathbf{e}^{j\theta_k} + l_\rho\mathbf{e}^{j\theta_\rho} = l_R\mathbf{e}^{j\theta_P} \tag{9}$$

$$l_{Hi}\mathbf{e}^{j\theta_{Hi}} + l_{ai}\mathbf{e}^{j\theta_{ai}} = l_v\mathbf{e}^{j\theta_v} + l_{ci}\mathbf{e}^{j\theta_{ci}} \tag{10}$$

where l_{Hi}, l_{ai}, and l_R are the length of the carrier, eccentricity of the crankshafts, and pitch circle radius of the pin gear, respectively. $l_{ci} = l_{Hi}$ and $l_v = l_{ai}$,l_{b1},l_{b2},l_n,l_ρ, and l_k are the lengths of the virtual linkages, respectively. θ_v is the rotation angle of the virtual crankshaft.

By the derivation of Eqs. (8)–(10) with respect to time, and yields

$$j\dot{\theta}_{b1}l_{b1}\mathbf{e}^{j\theta_{b1}} + j\dot{\theta}_n l_n\mathbf{e}^{j\theta_n} = j\dot{\theta}_{H1}l_{H1}\mathbf{e}^{j\theta_{H1}} + j\dot{\theta}_{b2}l_{b2}\mathbf{e}^{j\theta_{b2}} \tag{11}$$

$$j\dot{\theta}_v l_v\mathbf{e}^{j\theta_v} + j\dot{\theta}_k l_k\mathbf{e}^{j\theta_k} + j\dot{\theta}_\rho l_\rho\mathbf{e}^{j\theta_\rho} = j\dot{\theta}_p l_R\mathbf{e}^{j\theta_p} \tag{12}$$

$$j\dot{\theta}_{Hi}l_{Hi}\mathbf{e}^{j\theta_{Hi}} + j\dot{\theta}_{ai}l_{ai}\mathbf{e}^{j\theta_{ai}} = j\dot{\theta}_v l_v\mathbf{e}^{j\theta_v} + j\dot{\theta}_{ci}l_{ci}\mathbf{e}^{j\theta_{ci}} \tag{13}$$

Taking the dot product on both sides of Eq. (11) with $\boldsymbol{e}^{-j\theta n}$, Eq. (12) with $\boldsymbol{e}^{-j\theta\rho}$, Eq. (13) with $\boldsymbol{e}^{-j\theta ai}$, and Eq. (13) with $\boldsymbol{e}^{-j\theta ci}$, respectively, and then velocity equation of each loop is derived as

$$l_{b1}\dot{\theta}_{b1}\sin\theta_{b1,n} = l_{H1}\dot{\theta}_{H1}\sin\theta_{H1,n} + l_{b2}\dot{\theta}_{b2}\sin\theta_{b2,n} \tag{14}$$

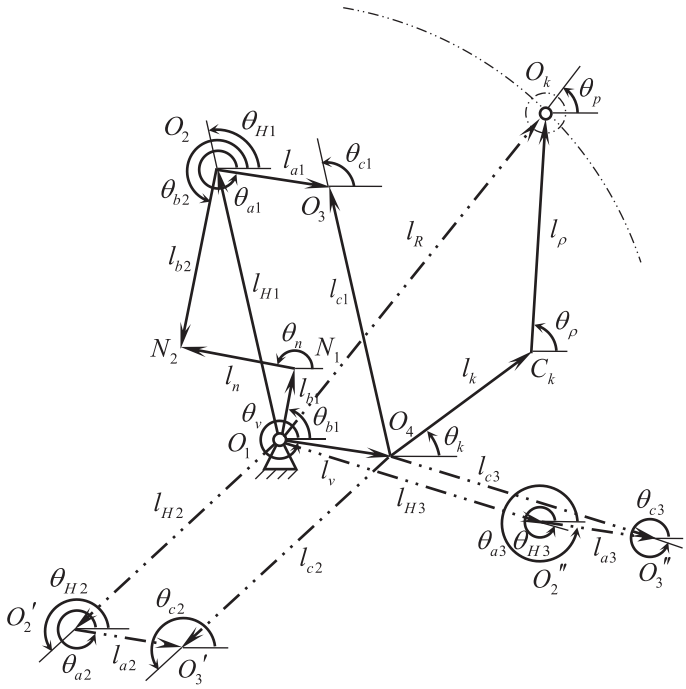


Fig. 5. Closed-loop vector model of the mechanism.

$$l_v\dot{\theta}_v \sin\theta_{v,\rho} + l_k\dot{\theta}_k \sin\theta_{k,\rho} = l_R\dot{\theta}_p \sin\theta_{p,\rho} \tag{15}$$

$$l_{Hi}\dot{\theta}_{Hi} \sin\theta_{Hi,ai} = l_v\dot{\theta}_v \sin\theta_{v,ai} + l_{ci}\dot{\theta}_{ci} \sin\theta_{ci,ai} \tag{16}$$

$$l_{Hi}\dot{\theta}_{Hi} \sin\theta_{Hi,ci} + l_{ai}\dot{\theta}_{ai} \sin\theta_{ai,ci} = l_v\dot{\theta}_v \sin\theta_{v,ci} \tag{17}$$

where $\theta_{x,y} = \theta_x - \theta_y$.

When all components are in ideal dimensions without errors, the equivalent virtual linkages satisfy

$$\dot{\theta}_{b1} = \dot{\theta}_{b2},\ \dot{\theta}_{Hi} = \dot{\theta}_{ci},\ \dot{\theta}_{ai} = \dot{\theta}_v \tag{18}$$

Substituting Eq. (18) into Eqs. (14)–(17) results in the speed ratio of each loop

$$g_1 = \frac{\dot{\theta}_{b2}}{\dot{\theta}_{H1}} = \frac{l_{H1}\sin\theta_{H1,n}}{l_{b2}\sin\theta_{b2,n}} = -\frac{z_1 + z_2}{z_2} \tag{19}$$

$$g_2 = \frac{\dot{\theta}_v}{\dot{\theta}_{ai}} = \frac{l_{ai}\sin\theta_{ai,ci}}{l_v\sin\theta_{v,ci}} = 1 \tag{20}$$

$$g_3 = \frac{\dot{\theta}_k}{\dot{\theta}_v} = -\frac{l_v\sin\theta_{v,\rho}}{l_k\sin\theta_{k,\rho}} = -\frac{z_5 - z_4}{z_4} \tag{21}$$

$$g_4 = \frac{\dot{\theta}_{Hi}}{\dot{\theta}_{ci}} = \frac{l_{ci}\sin\theta_{ci,ai}}{l_{Hi}\sin\theta_{Hi,ai}} = 1 \tag{22}$$

where g_1 and g_3 denote speed ratios of the planetary gear to carrier and the cycloidal gear to crankshafts, respectively. g_2 and g_4 represent speed ratios of the opposite sides of the parallelograms, respectively.

4. Rotational transmission error model

According to the closed-loop vector model of the mechanism, the analytical model of the RTE of the gear train will be established with the aid of loop incremental method in this section.

4.1. Analytical model

Solving the total differential of Eqs. (8)–(10) with respect to each parameter, and then taking the dot product on both sides of Eq. (8) with $\boldsymbol{e}^{-j\theta n}$, Eq. (9) with $\boldsymbol{e}^{-j\theta\rho}$, Eq. (10) with $\boldsymbol{e}^{-j\theta ai}$, Eq. (10) with $\boldsymbol{e}^{-j\theta ci}$, respectively. The following equations are consequently obtained

$$\Delta l_{b1}\cos\theta_{b1,n} - \Delta\theta_{b1}l_{b1}\sin\theta_{b1,n} + \Delta l_n = \Delta l_{H1}\cos\theta_{H1,n} - \Delta\theta_{H1}l_{H1}\sin\theta_{H1,n} + \Delta l_{b2}\cos\theta_{b2,n} - \Delta\theta_{b2}l_{b2}\sin\theta_{b2,n} \tag{23}$$

$$\Delta l_v\cos\theta_{v,\rho} - \Delta\theta_v l_v\sin\theta_{v,\rho} + \Delta l_k\cos\theta_{k,\rho} - \Delta\theta_k l_k\sin\theta_{k,\rho} + \Delta l_\rho = \Delta l_R\cos\theta_{p,\rho} - \Delta\theta_p l_R\sin\theta_{p,\rho} \tag{24}$$

$$\Delta l_{Hi}\cos\theta_{Hi,ai} - \Delta\theta_{Hi}l_{Hi}\sin\theta_{Hi,ai} + \Delta l_{ai} = \Delta l_v\cos\theta_{v,ai} - \Delta\theta_v l_v\sin\theta_{v,ai} + \Delta l_{ci}\cos\theta_{ci,ai} - \Delta\theta_{ci}l_{ci}\sin\theta_{ci,ai} \tag{25}$$

$$\Delta l_{Hi}\cos\theta_{Hi,ci} - \Delta\theta_{Hi}l_{Hi}\sin\theta_{Hi,ci} + \Delta l_{ai}\cos\theta_{ai,ci} - \Delta\theta_{ai}l_{ai}\sin\theta_{ai,ci} = \Delta l_v\cos\theta_{v,ci} - \Delta\theta_v l_v\sin\theta_{v,ci} + \Delta l_{ci} \tag{26}$$

where Δl_x and $\Delta\theta_x$ refer to the length error and rotation angle error of the corresponding linkage l_x, respectively.

Note that $l_{b2}\boldsymbol{e}^{j\theta b2}$ and $l_{ai}\boldsymbol{e}^{j\theta ai}$ respectively indicate the planetary gear and crankshafts, and they are fixed together as one component, hence

$$\Delta\theta_{b2} = \Delta\theta_{ai} \tag{27}$$

Similarly, both $l_{ci}\boldsymbol{e}^{j\theta ci}$ and $l_k\boldsymbol{e}^{j\theta k}$ stand for the cycloidal gear, thus

$$\Delta\theta_{ci} = \Delta\theta_k \tag{28}$$

Substituting Eqs. (19)–(22), (27), and (28) into Eqs. (23)–(26), the analytical RTE expression can be written as

$$\Delta\theta_H = \frac{g_2g_3g_4}{1+g_1g_2g_3g_4}f_1 + \frac{g_3g_4}{1+g_1g_2g_3g_4}\frac{1}{3}\sum_{i=1}^{3}f_{2i} + \frac{g_4}{1+g_1g_2g_3g_4}f_3 + \frac{1}{1+g_1g_2g_3g_4}\frac{1}{3}\sum_{i=1}^{3}f_{4i} \tag{29}$$

It can be seen that g_j (j = 1–4) actually indicates error transfer coefficient for each subsystem.

Suppose that the cycloid-pin gear drive is one-tooth difference, substituting $z_5 - z_4 = 1$ into Eq. (29), and then the analytical RTE expression of RV reducer is further rewritten as

$$\Delta\theta_H = \frac{-\frac{1}{z_4}f_1 - \frac{1}{3z_4}\sum_{i=1}^{3}f_{2i} + f_3 + \frac{1}{3}\sum_{i=1}^{3}f_{4i}}{1+\frac{z_1+z_2}{z_2z_4}} \tag{30}$$

where

$$f_1 = \frac{1}{\sin\theta_{b2,n}}\left(-\frac{\Delta l_{b1}}{l_{b2}}\cos\theta_{b1,n} + \frac{l_{b1}}{l_{b2}}\Delta\theta_{b1}\sin\theta_{b1,n} - \frac{\Delta l_n}{l_{b2}} + \frac{\Delta l_{H1}}{l_{b2}}\cos\theta_{H1,n} + \frac{\Delta l_{b2}}{l_{b2}}\cos\theta_{b2,n}\right) \tag{31}$$

$$f_{2i} = \frac{1}{\sin\theta_{v,ci}}\left(-\frac{\Delta l_{Hi}}{l_v}\cos\theta_{Hi,ci} - \frac{\Delta l_{ai}}{l_v}\cos\theta_{ai,ci} + \frac{\Delta l_v}{l_v}\cos\theta_{v,ci} + \frac{\Delta l_{ci}}{l_v}\right) \tag{32}$$

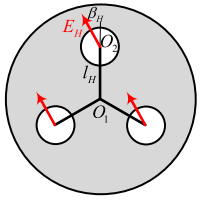
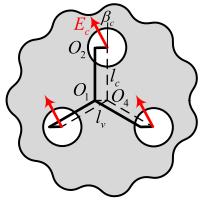
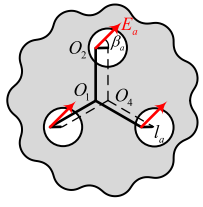
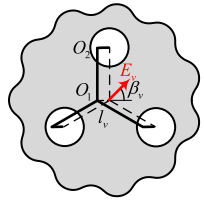
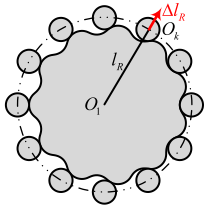
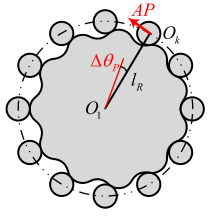
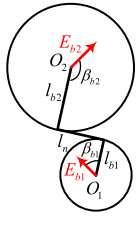
$$f_3 = \frac{1}{\sin\theta_{k,\rho}}\left(\frac{\Delta l_v}{l_k}\cos\theta_{v,\rho} + \frac{\Delta l_k}{l_k}\cos\theta_{k,\rho} + \frac{\Delta l_\rho}{l_k} - \frac{\Delta l_R}{l_k}\cos\theta_{p,\rho} + \frac{l_R}{l_k}\Delta\theta_p\sin\theta_{p,\rho}\right) \tag{33}$$

$$f_{4i} = \frac{1}{\sin\theta_{Hi,ai}}\left(\frac{\Delta l_{Hi}}{l_{Hi}}\cos\theta_{Hi,ai} + \frac{\Delta l_{ai}}{l_{Hi}} - \frac{\Delta l_v}{l_{Hi}}\cos\theta_{v,ai} - \frac{\Delta l_{ci}}{l_{Hi}}\cos\theta_{ci,ai}\right) \tag{34}$$

where f_1 and f_3 are the RTE caused by original errors of the involute and cycloid-pin gear transmission subsystems, respectively. f_{2i} and f_{4i} are the RTE resulted by original errors of the crankshafts when they work as the input and output subsystems, respectively.

Eq. (29) embodies the core merits of this article, and it is a universal model with explicit equations to reveal relations between the original error of each component and the RTE of entire machine. Hence, it provides crucial technical support for precision analysis and design of the key components.

Table 1
Manufacturing errors.

Error source	Position deviation of holes on output disc (E_H,β_H)	Position deviation of holes on cycloidal gear (E_c,β_c)	Eccentricity error of crankshafts (E_a,β_a)	Centre error of cycloidal gear (E_v, β_v)
Calculation method	$\Delta l_H = E_H\cos\beta_H$	$\Delta l_c = E_c\cos\beta_c$	$\Delta l_a = E_a\cos\beta_a$	$\Delta l_v = E_v\cos\beta_v$
Schematic diagram				
Error source	Pitch circle radius error of pin gear	Accumulative pitch error of pins AP	Geometrical eccentric errors of pitch circles of sun gear (E_{b1}, β_{b1}) and planetary gear (E_{b2}, β_{b2})	Errors of cycloidal tooth profile δ and pin radius Δr
Calculation method	Δl_R	$\Delta\theta_p = AP/l_R$	$\Delta l_{b1} = E_{b1}\cos\beta_{b1}$, $\Delta l_{b2} = E_{b2}\cos\beta_{b2}$, $\Delta l_n = E_{b1}\cos(\theta_{b1}+\beta_{b1}-\theta_n) + E_{b2}\cos(\theta_{b2}+\beta_{b2}-\theta_n)$	Δl_k and Δl_ρ
Schematic diagram				Refer to Fig. 6

Note: E_x is the error value, and β_x is the angle between the error direction and reference direction.

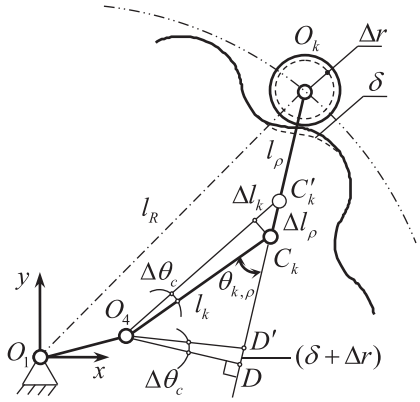


Fig. 6. Errors of cycloidal tooth profile and pin radius.

4.2. Original errors

Through analysing Eqs. (31)–(34), the original errors are all denoted by the linkage length errors Δl_x, except for the angle error of input shaft $\Delta\theta_{b1}$. Accordingly, it is essential to research the corresponding equivalent errors and calculation methods of the manufacturing and assembly errors of the reducer.

Manufacturing errors include those of crankshaft eccentricity, hole deviation, cycloidal tooth profile, pin radius, pitch circles, etc., and their equivalent calculation methods are listed in Table 1.

Referring to Fig. 6, suppose that the errors of cycloidal tooth profile and pin radius are δ and Δr, respectively. The cycloidal gear needs to rotate a small angle $\Delta\theta_c$ to contact with the k-th pin, and it moves a tiny displacement DD'. $\Delta\theta_c$ is thus expressed as

$$\Delta\theta_c = \frac{\delta + \Delta r}{l_k \sin\theta_{k,\rho}} \tag{35}$$

Meanwhile, the curvature centre of the tooth profile moves from C_k to $C_k{}'$, and the length errors of the virtual linkages l_k and l_ρ are represented as

$$\begin{cases} \Delta l_k = (-1)^n \frac{(\delta+\Delta r)}{\sin^2\theta_{k,\rho}} \cos\theta_{k,\rho} \\ \Delta l_\rho = (-1)^{n+1} \frac{(\delta+\Delta r)}{\sin^2\theta_{k,\rho}} \end{cases} (n = 0 \text{if} \rho_c \geq 0, n = 1 \text{if} \rho_c < 0) \tag{36}$$

where ρ_c is the curvature radius of the contact point.

Assembly errors involve those of involute/cycloidal gears and carrier, crankshafts and output disc, uneven distribution of the gears, etc., and they can be calculated by the similar methods. For instance, the assembly error of output disc (A_o, γ_o) is expressed as

$$\begin{cases} \Delta l_H = -A_o \cos(\gamma_o - \theta_H) \\ \Delta l_v = -A_o \cos(\gamma_o - \theta_v) \end{cases} \tag{37}$$

Based on the above analysis, it is worth mentioning that one linkage length error in the model virtually can represent multiple original errors. For example, Δl_H indicates the assembly error of carrier and position deviation of holes on the output disc. Also, Δl_ρ and Δl_{b2} severally represent the cycloidal tooth profile error and geometrical eccentric error of planetary gear pitch circle, and they denote the assembly errors of the gears as well. Consequently, the original errors of one component can be calculated with the maximum allowance in the model.

5. Validation of the model

The RTE test will be carried out to verify effectiveness of the presented model in comparison with the numerical simulation results, and the influence investigation of original errors is implemented on the RTE of entire machine.

5.1. Comparison with Blanche's model

The influence of the cycloidal tooth profile error on the RTE is simulated by Eqs. (30) and (36) with relevant parameters listed in Table 2, and the results are compared with those of Blanche's model [1], as shown in Fig. 7. The agreement between the two models at their results proves the feasibility of the equivalent calculation method of cycloidal tooth profile error.

Table 2
Parameters of Blanche's example.

Tooth number of cycloidal gear z_4	Tooth number of pin gear z_5	Pitch circle radius of pin gear l_R	Short width coefficient k_1	Error of cycloidal tooth profile δ
20	21	50.8mm	0.6	0.00254mm

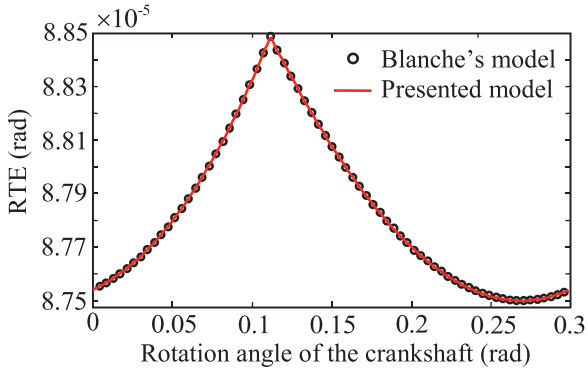
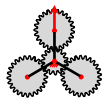
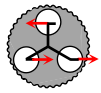


Fig. 7. Comparison with Blanche's model.

Table 3
Basic parameters of the RV-80E prototypes.

Parameter	Value	Parameter	Value
Tooth number of sun gear z_1	10	Module m (mm)	1.75
Tooth number of planetary gear z_2	38	Working pressure angle α' (°)	20
Tooth number of pin gear z_5	40	Pin radius r (mm)	4
Pitch circle radius of pin gear l_R (mm)	77.5	Crankshaft eccentricity l_a (mm)	1.5

Table 4
Original errors of the two protypes.

Error	Error value (mm)		Error direction	
	prototype #1	prototype #2		
E_{b2}	0.005	0.005		$\beta_{b2} = 180°$
E_a	0.003	0.002		$\beta_a = 180°$
E_H	0.004	0.005		$\beta_H = 180°$
E_c	0.005	0.005		$\beta_c = 0°$
δ	0.005	0.004	Normal direction of cycloidal tooth profile	
Δr	0.002	0.002	Inner equidistance of pin radius	
Δl_R	0.005	0.004	Inner equidistance of pitch circle of pin gear	
AP	$0.005\sin(2k\pi/z_5)$	$0.005\sin(2k\pi/z_5)$	Tangential direction of pitch circle of pin gear	

5.2. Comparison with test results

The RTE test is conducted on two prototypes whose basic parameters are listed in Table 3, and the original errors considered in the model are displayed in Table 4. some manufacturing error values are determined by the machining tolerance,

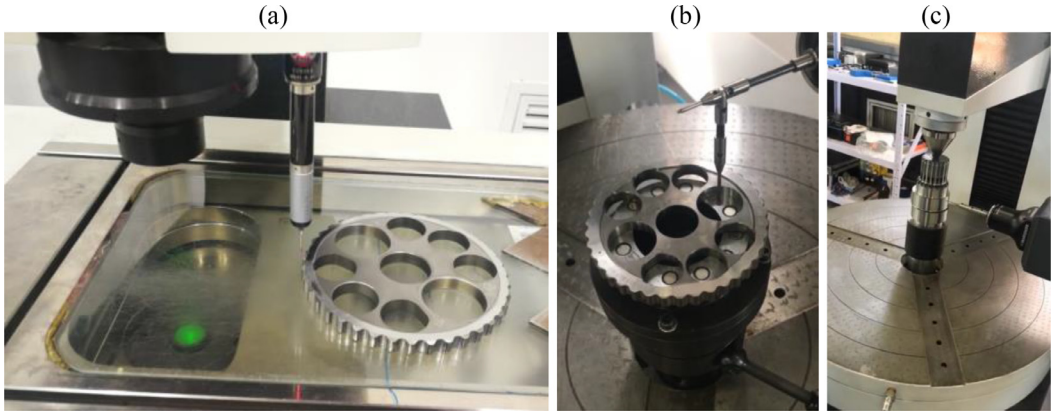


Fig. 8. Component precision measurement, (a) cycloidal tooth profile, (b) shaft holes on cycloidal gear, and (c) crankshaft eccentricity.

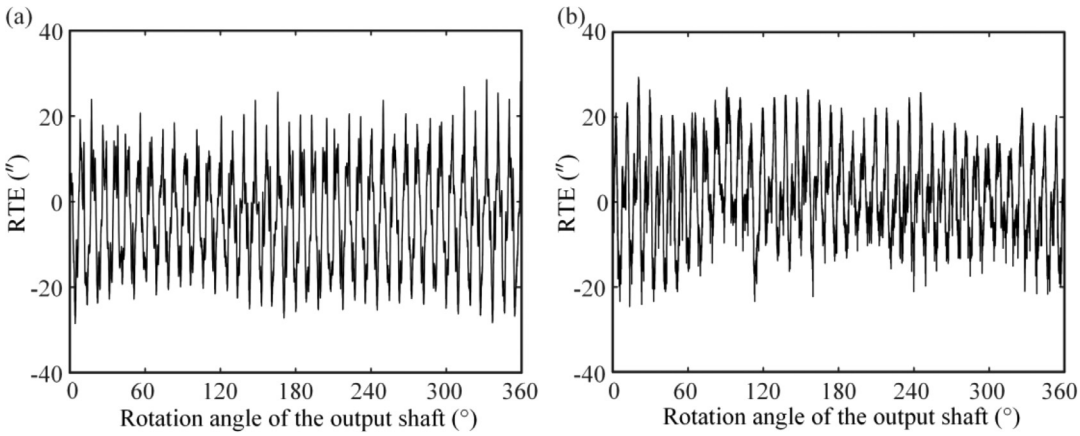


Fig. 9. time-domain curves of RTE of prototype #1, (a) simulation result and (b) test result.

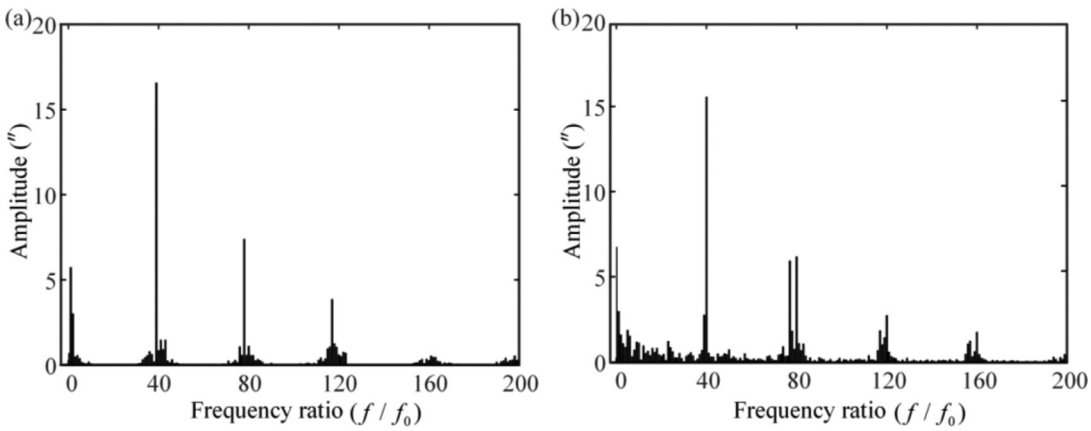


Fig. 10. frequency-domain curves of RTE of prototype #1, (a) simulation result and (b) test result.

and other ones, such as cycloid tooth profile, hole position, crankshaft eccentricity, are precisely measured by technical instruments before the prototype assembly, as shown in Fig. 8.

The time-domain and frequency-domain curves of simulation and test results of the prototype #1 are respectively illustrated in Figs. 9 and 10, and those of the prototype #2 are respectively illustrated in Figs. 11 and 12. In Figs. 10 and 12, f and f_0 respectively denote the frequency of the output error and the rotation frequency of the output shaft, and f/f_0 is set as the abscissa to eliminate their dimensions.

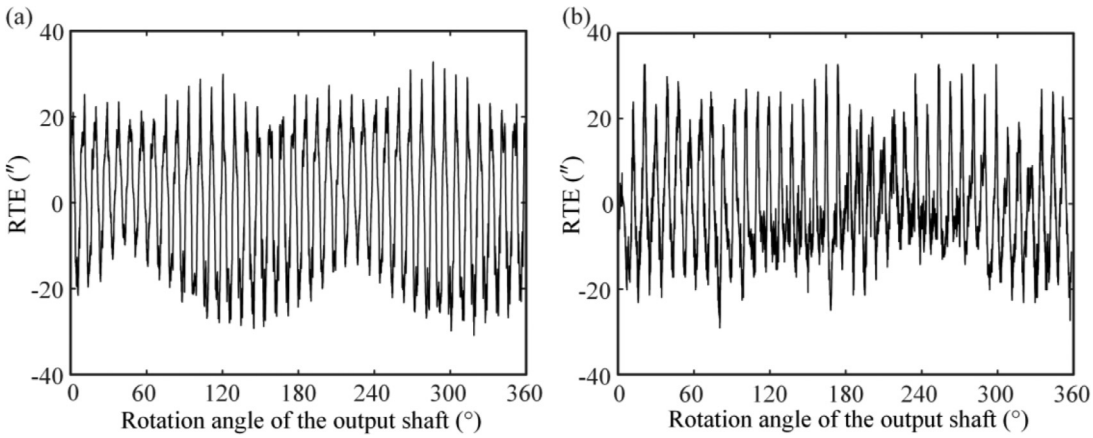


Fig. 11. time-domain curves of RTE of prototype #2, (a) simulation result and (b) test result.

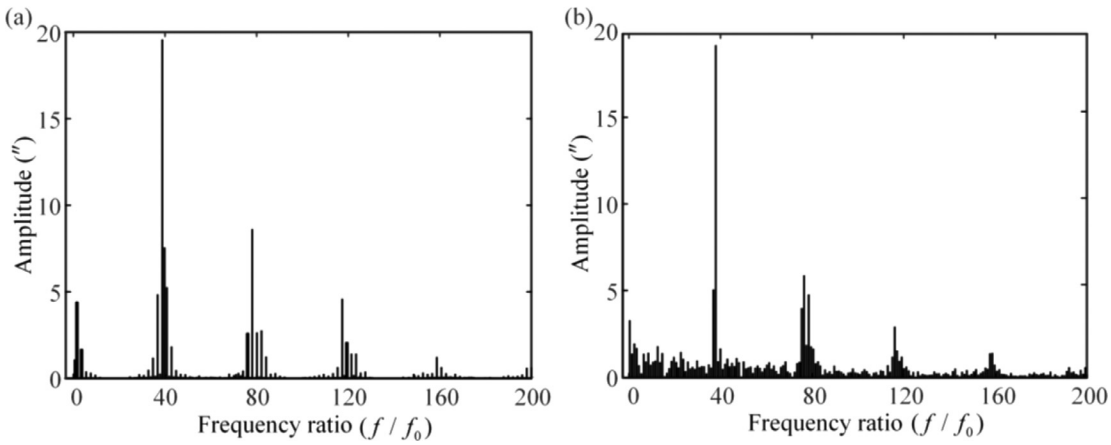


Fig. 12. frequency-domain curves of RTE of prototype #2, (a) simulation result and (b) test result.

Figs. 9(a) and 9(b) show that the RTE fluctuation of the simulation and test results of the prototype #1 are −28.83″~27.32″ and −29.68″~30.39″, respectively, and it can be seen from Figs. 11(a) and 11(b) that those of the prototype #2 are −32.16″~32.87″ and −31.51″~33.13″, respectively. The simulation and test results of the two prototypes are both in good agreement, respectively.

It can be observed from Figs. 10 and 12 that the RTE all have several frequency components of 40,80,120,160, etc., which behave as multiples of the meshing frequency of cycloidal gear and pins, and the maximum amplitudes all occur at 40, indicating that cycloidal stage errors have significant influence on the RTE. Moreover, the frequency component of 1 is caused by the position deviation of holes on output disc, since the carrier rotates a circle corresponding to one rotation of the output shaft.

In conclusion, both the fluctuation amplitudes of the time-domain curves and frequency components of the frequency-domain curves are consistent between the simulation and test results of the two prototypes, which proves the effectiveness of the proposed model.

The schematic diagram and hardware design of the test bench are illustrated in Fig. 13, including servo motor 1, torque sensor 2, input angle encoder 3, prototype 4, output angle encoder 5, and they are installed on the test base successively with the same installation height. The relevant parameters of the instruments are listed in Table 5.

The test is performed under low-speed and no-load conditions, and the servo motor drives the prototype to rotate at input speed of 40.0 rpm with 3% of rated torque. Each angle encoder is separately connected to the input and output shafts through flanges, and they collect the angle values (θ_1 and θ_H) corresponding to 11,440 test points within one rotation of the output shaft. Suppose that the speed ratio of the prototype is denoted as I, and then the test result is calculated by

$$\Delta\theta_H = \theta_H - \frac{\theta_1}{I} \tag{38}$$

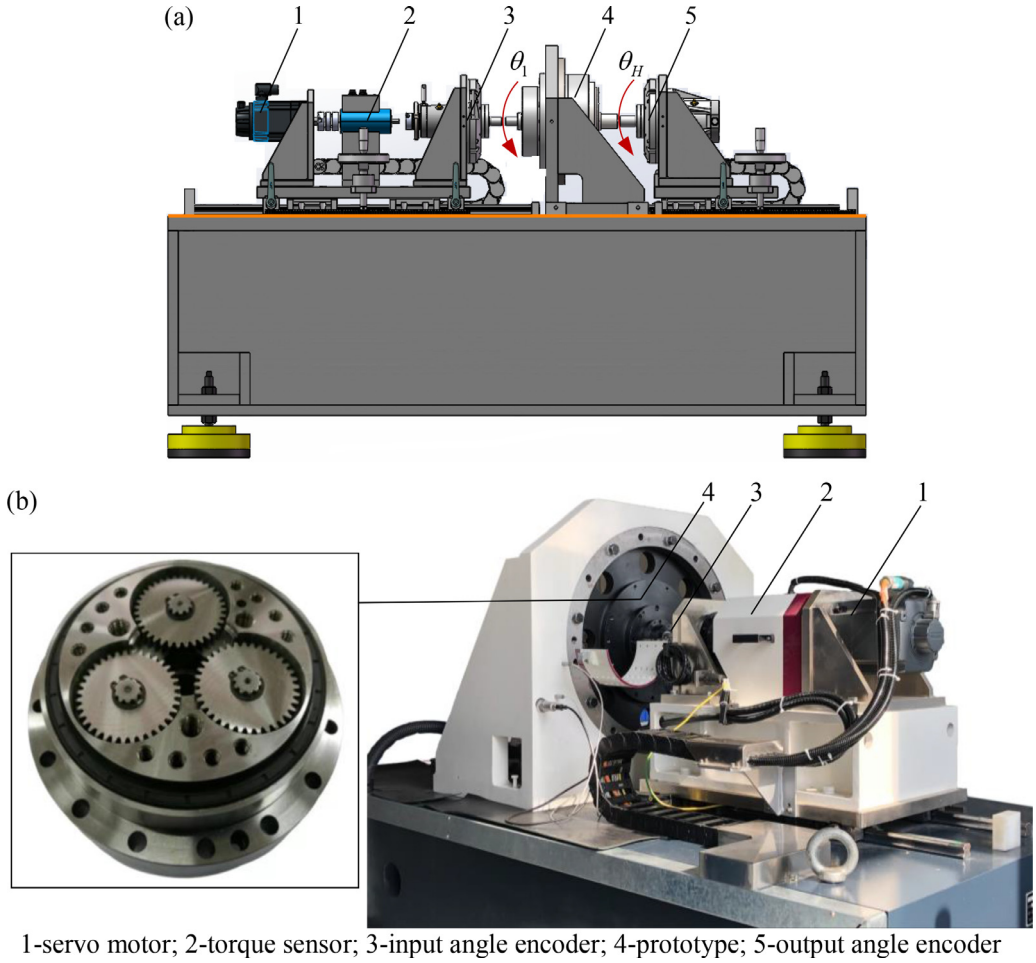


Fig. 13. Test bench, (a) schematic diagram and (b) hardware design.

Table 5
Relevant parameters of instruments.

Servo motor SIEMENS	Angle encoder HEIDENHAIN RON785	Torque sensor KISTLER 4503B
Rated speed 3000rpm	Line number 18,000	Speed test accuracy ± 0.1rpm
Maximum speed 6000rpm	System accuracy ± 2″	Torque test accuracy ± 0.05%F.S
Rated power 2.73kW	Single signal period error ≤ 0.7″	Double-range Measuring 1Nm/10Nm
Rated torque 8.7Nm	Allowable centre runout ± 0.1mm	Maximum speed 20000rpm
	Allowable mechanical speed 1000rpm	Output signal range 100 kHz ± 40kHz

Figs. 9 and 11 demonstrate comprehensive effect of the original errors on the RTE, then in order to research the influence of individual error on the RTE, eight original errors of the prototype #1 listed in Table 4 are respectively simulated, as illustrated in Figs. 14(a) and 14(b).

The outcomes indicate that the original errors of the low-speed stage, including δ, AP, E_c,E_a,E_H,Δl_R,Δr, have greater impacts on the RTE than that of the high-speed stage E_{b2}. It can be seen that variations of E_{b2},E_c,E_a, and E_H are in identical high-frequency range corresponding to planetary gear rotation and cycloidal gear revolution, respectively, while the variation of AP is in low-frequency range owing to the cycloidal gear rotation. Also, variations of E_c and E_a exhibit relatively large amplitude as well due to the influence on both rotation and revolution of the cycloidal gear. In addition, tiny fluctuations actually occur in high-frequency range when the cycloidal tooth profile and pin radius have errors, and variations of δ and Δl_R approximately appear to be a horizontal straight line.

Moreover, δ is taken as an example to analyse the effect of feedback error on the RTE, and the results are shown in Fig. 14(c). It can be known that the feedback error accounts for about 3% in this case, and such error can be neglected under the allowable engineering accuracy requirements.

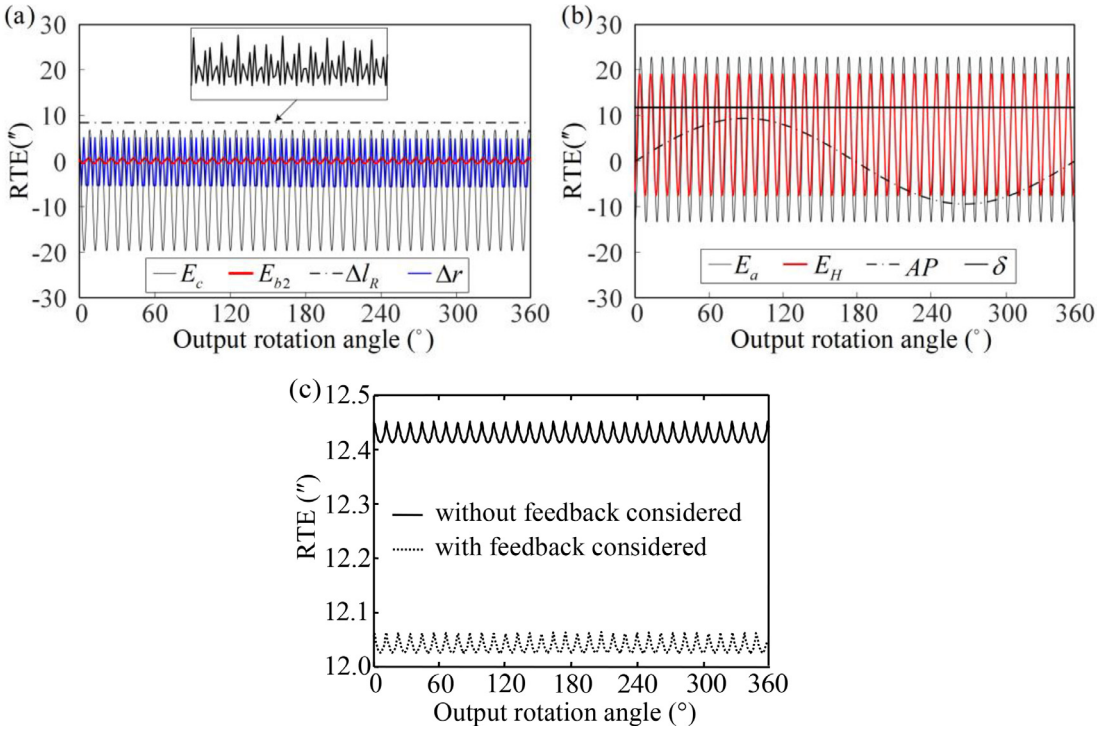


Fig. 14. Variation of RTE versus the output rotation angle, (a) influence of E_c,E_{b2}, Δl_R, Δr on RTE, (b) influence of E_a,E_H, AP, δ on RTE, and (c) influence of feedback error.

6. Conclusions

This paper presents a modelling approach for the kinematic equivalent mechanism and rotational transmission error. The conclusions are drawn as follows:

(1) According to the structural characteristics that the rotation and revolution of the cycloidal gear share three crankshafts, the instantaneous kinematic equivalent mechanism is established by means of lower pairs replacing higher pairs, and the error modelling of entire machine is thus converted into the error analysis of equivalent multi-loop mechanism. It offers a new idea for the and error modelling of RV reducer and other gear transmission systems.

(2) Based on the proposed kinematic equivalent mechanism, the analytical model of the rotational transmission error of RV reducer is developed with the aid of loop incremental method, which employs explicit equations to reveal influence of original errors of high-speed and low-speed stages on the transmission precision. It provides critical technical support for the accuracy analysis and design of the key components of RV reducer.

Declaration of Competing Interest

The authors declare no competing financial interests, and they have no conflicts of interest to this work.

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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